



Drug dosage forms

Pharmacology Department
Faculty of Medicine



Objectives

Upon completion of this lesson, the student will be able to:

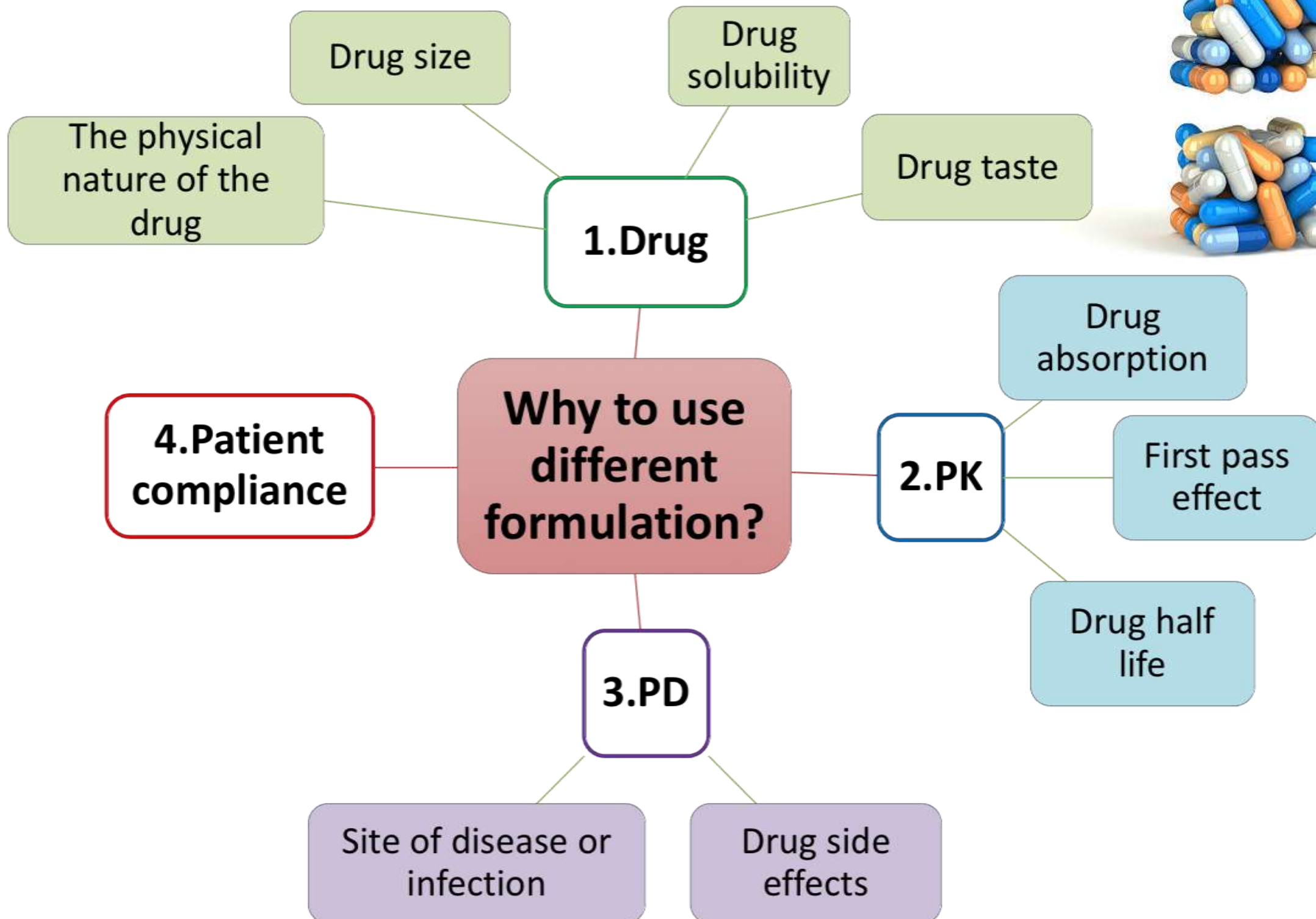


Differentiate between drug dosage forms.

List examples for each dosage form.

Explain suitable route for each drug form.

Why to use different formulation?



Drug dosage forms

Solid dosage forms

Tablets

Capsules

Lozenges

Powders

Pills

Liquid dosage forms

Semi-solid dosage forms

Others

Nasal

Inhalational



Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

Powders

Pills

I-Solid dosage forms

- Capsules, sustained-release capsules, tablets, pills
- They are used to divide a drug or mix drugs into **definite doses** and avoid the inconvenience of preparing the dose from dry powders.
- Tablets are a convenient way of giving drugs that have an **unpleasant taste**



Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

Powders

Pills

Capsules

- They dissolve readily in the stomach and make the contents available for absorption only slightly less quickly than a liquid medicament.
- Are usually made of **gelatine** and may be hard or soft.
 - Hard gelatine capsule : contains powder
 - Soft gelatine capsule : contains Liquids



- Gelatine capsules may be coated with a substance that resists the action of gastric juice and so will not disintegrate until they reach the alkaline secretions of the intestine. Such capsules are said to be '**enteric coated**'.

Enteric-Coated

Non-Coated

Resistant to gastric acidic fluid

60-80% intestinal absorption

Acidic stomach destroys 80% of placenta ingredients

20% available for absorption in intestine



Drug dosage forms

Solid dosage forms

Capsules

Tablets

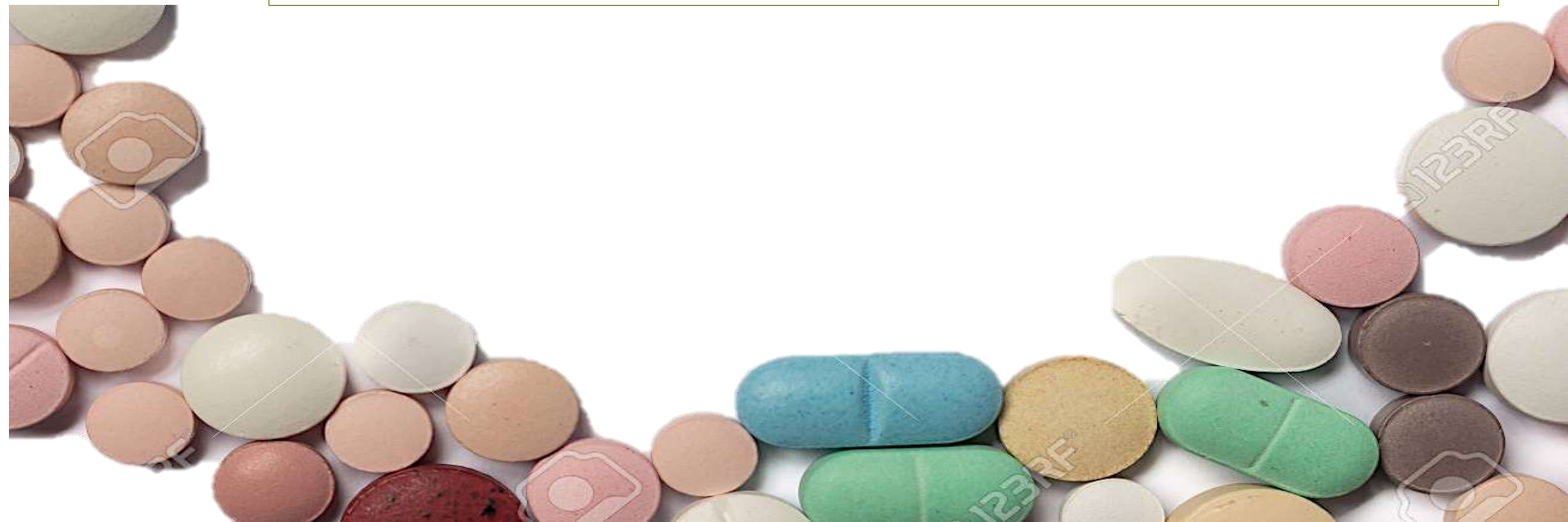
Lozenges

Powders

Pills

Tablets

- Tablets are preparations of powdered drug that are compressed or moulded into small disks.
- They may be made with or without a diluent (dextrose, lactose, starch).
- They may differ greatly in size, weight, and shape.



Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

Powders

Pills

Tablets

Tablets

Normal tablets



Film coated



Effervescent



Sugar coated



Chewable



Sustained release

Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

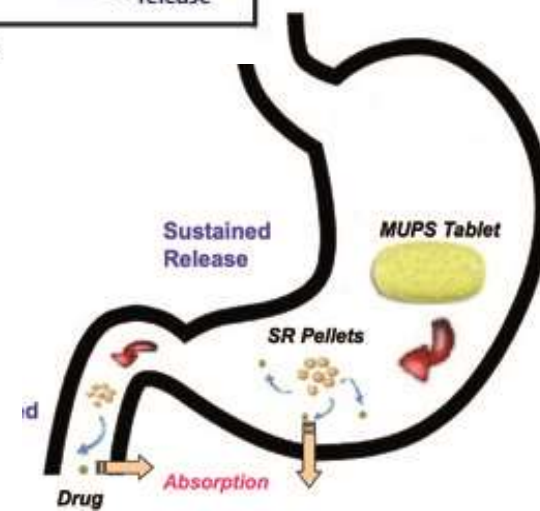
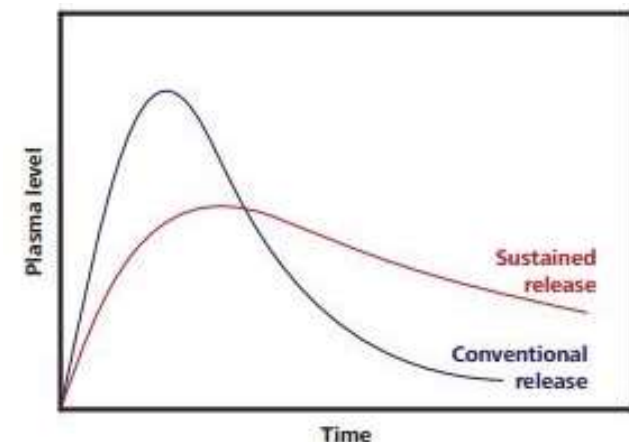
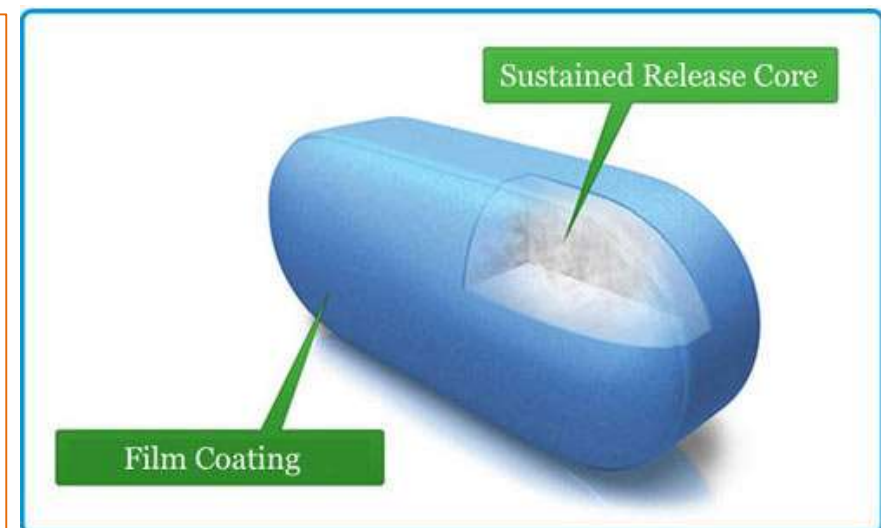
Powders

Pills

Tablets

Sustained-release dosage forms

- Dosage forms providing for **gradual but continued** release of drug (longer duration of action and can **NOT crush it**).
- Sustained-release dosage forms contain small particles of the drug coated with materials that require a **varying** amount of time to dissolve.
- Some particles dissolve and are absorbed almost immediately, others require 2 or 3 hours.
- This provides for a **long continuous** period of absorption and effect.



Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

Powders

Pills

Lozenges

- Troches or lozenges are flat, round, or rectangular preparations that are **held in the mouth until they dissolve**.
- They usually contain water, sugar, and gelatin.
- They temporarily produce a high concentration of the drug in the oral cavity.



Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

Powders

Pills

Pills

- Pills are mixtures of a drug or drugs with some cohesive materials.
- The mass is molded into globular, oval, or flattened bodies convenient for swallowing.
- They have been replaced to a great extent by capsules and tablets



Contraceptive Pills

Drug dosage forms

Solid dosage forms

Capsules

Tablets

Lozenges

Powders

Pills

Powders

➤ Powders are finely divided solid drugs or mixtures of drugs for internal or external use.

- ❖ Dusting powders
- ❖ Granules
- ❖ Sachets



Drug dosage forms

II-Liquid dosage forms

Liquid dosage forms



Drug dosage forms

II-Liquid dosage forms

Liquid dosage forms

Solutions

- Aqueous solutions have one or more substances dissolved in water.
- Examples are:
 - ❖ strong iodine solution.
 - ❖ (Lugol's Solution), epinephrine nasal solution.
- **Syrups:** e.g Promethazine- codeine (cough syrup).



Syrups tend to be thicker than water.



Drug dosage forms

II-Liquid dosage forms

Liquid dosage forms

Suspensions

are defined as preparations of finely divided drugs either intended for suspension or already in suspension in some suitable liquid vehicle .



Drug dosage forms

II-Liquid dosage forms

Liquid dosage forms

Emulsions

- are suspensions of fats or oils in water with the aid of an, emulsifying agent.
- An example is cod liver oil emulsion.



Drug dosage forms

II-Liquid dosage forms

Liquid dosage forms

Elixirs

are aromatic, sweetened, alcoholic preparations, frequently used as flavored vehicles, such as aromatic elixir, or as active medicinal agents if they are medicated elixirs, such as phenobarbital elixir.



Drug dosage forms

Liquid dosage forms

II-Liquid dosage forms

Drops

eye (ophthalmic) – ear
(otic) – nose (Nasal) – oral
(commonly for infants)



Lotions



Sprays

- nasal – oral - local



II-Liquid dosage forms

Liquid dosage forms

Ampoules and
the like

- 1-Ampoules (Single Dose).
- 2- vials (multi -dose containers with rubber capped stopper).
- 3- Bottles.
- 4- Disposable syringes.



Drug dosage forms

Semi-solid dosage forms

III-Semi-solid dosage forms

1- Ointments and the like

- Ointments
- Creams
- Gels
- Pastes



2- Transdermal therapeutic system



3- Suppositories and the like

- Rectal supp.
- Vaginal inserts (tablets – ovules – supp. pessary)



Powder

Pastes

Shake lotions

Cream pastes
Cooling pastes

Greases:
Ointments

Liquids:
Gels
Solutions
Lotions

Water-in-oil (W/O)
emulsions:
ointments

Oil-in-water (O/W)
emulsions:
creams



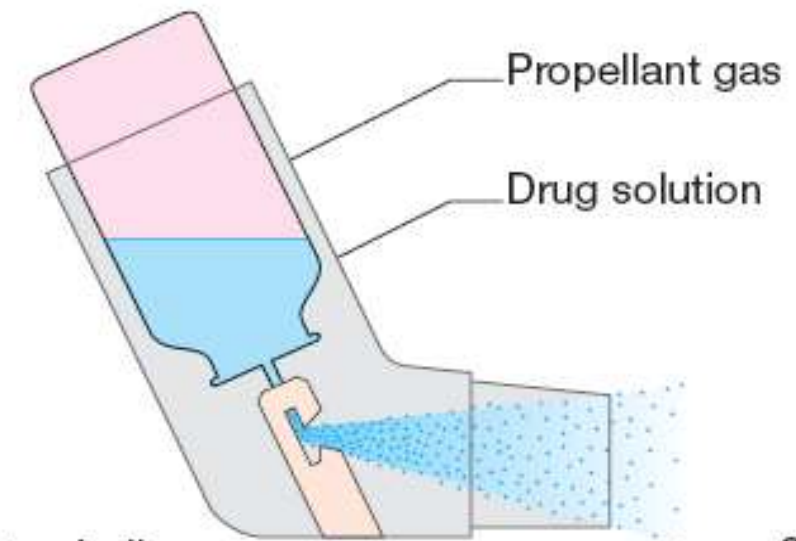
Drug dosage forms

Others

Nasal

Inhalational

IV- Other dosage forms



Jet nebulizer



Thank you